

M000 · Changing a Variable — English Worksheet (Beginner)

Topic: Changing a Variable While You Build · **Course:** Builder Basics · **Level:** Beginner · **Time:** about 30 minutes

1 · Predict 🧠

A variable can **change while the code runs**. `height = height + 1` means: add 1 to what `height` holds.

Read the steps. Circle your answer.

```
height = 1
repeat 3 times:
  place block down
  move up by height
  move forward
  height = height + 1
```

What does `height` hold on the first step? Circle one: 1 · 2 · 3

What does `height` hold on the second step? Circle one: 2 · 1 · 5

Does each step climb the same amount or more? Circle one: more each time · the same

2 · Find the Difference 🐛

Clean steps first, then a broken version. Circle the bug.

Pair A — Each step should climb higher than the one before.

```
# clean
height = 1
repeat 3 times:
    place block down
    move up by height
    move forward
    height = height + 1
```

```
# buggy
height = 1
repeat 3 times:
    place block down
    move up by height
    move forward
```

What is wrong? Circle one: `height` never changes · the loop is missing · a block is missing

What shape do you get from the buggy code? Circle one: every step the same · a staircase · a tower

Pair B — `height` should keep growing.

```
# clean
height = 1
repeat 3 times:
    place block down
    move up by height
    height = height + 1
```

```
# buggy
height = 1
repeat 3 times:
    height = 1
    place block down
    move up by height
    height = height + 1
```

What is wrong? Circle one: `height` goes back to 1 every time · `height` grows too fast · the loop runs too many times

3 · Fill the Gap

Each step should climb higher than the one before. One line is missing.

```
height = 1
repeat 3 times:
  place block down
  move up by height
  move forward
  _____
```

Which line is missing? Circle one: $\text{height} = \text{height} + 1$ · turn right · $\text{height} = 1$

4 · Show Your Work 📷 🎥

Switch to your homework world. Build a **staircase** where each step climbs higher than the one before. Then change the number you add each time and run it again.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does. Point at the line that changes your variable.

2 · Your build. Point the camera. Show both staircases.

Say these out loud, filling in the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Submit ✅

Send this worksheet + your video to teacher on KakaoTalk.